

Page	Description	Design
Home Page	Steel Industry - Home Subtitle: <i>Understanding the world's backbone material</i> Part A - What is Steel? Steel is an alloy predominately made of iron and carbon. It is metaphorically the backbone of the current society typically used in buildings, bridges, cars, ships, etc. Part B - How steel is made - Blast furnace basic oxygen furnace (BF-BOF) ~70% - Iron ores are reduced to iron as it enters the furnace. - Electric Arc Furnace (EAF) Route ~30% - They use electricity to melt the recycled steel. EAF is a faster and an environmentally friendly method. Part C - Global Snapshot 🌐 Global production: ~1.8 billion tonnes per year. 🇨🇳 China: ~929 Mt (~50% of global share). 🇰🇷 South Korea: ~58 Mt. Part D - Impact Dashboard 1 tonne of steel = ~1.89 tonnes of CO₂ The steel industry is responsible for 7~9% of global emissions. Steel Saved: XX tonnes this month Cost Savings: \$XXXXXXX Estimated CO₂ Saved: XXX tones	SteelOptimize Home Calculator How to Use Impact Steel Industry Understanding the world's backbone material What is Steel? Steel is an alloy predominately made of iron and carbon. It is metaphorically the backbone of the current society typically used in buildings, bridges, cars, ships, etc. How Steel is Made 70% Blast Furnace Basic Oxygen Furnace (BF-BOF) Iron ores are reduced to iron as it enters the furnace. 30% Electric Arc Furnace (EAF) Route They use electricity to melt the recycled steel. EAF is a faster and an environmentally friendly method. Global Snapshot 1.8B tonnes per year Global production 929 Mt ~50% global share China 58 Mt production South Korea Impact Dashboard 1 tonne of steel = ~1.89 tonnes of CO₂ The steel industry is responsible for 7~9% of global emissions XX tonnes Steel saved this month \$XXXXXXX Cost Savings XXX tonnes Estimated CO₂ Saved

How to Use

Title: How to Use the Calculator

- 1. Prepare your inputs
 - a. Width, desired length, number of steel
- 2. Run the Optimization
- 3. Review the results and save the inputs & outputs to a PDF/CSV
- 4. The remaining amount will be stored in the 'remainder' section.

**There is an option to add additional rows.



Calculator

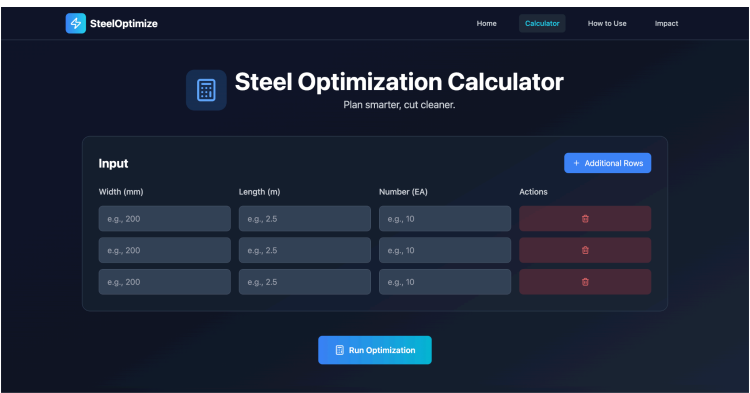
Header
Title: Steel Optimization Calculator
Subtitle (description): plan smarter, cut cleaner.

Part 1 - Input (Header)

width(mm)	length(m)	number(EA)

+ Additional Rows

Part 2 - Output Panel



	width(mm)length(m)number(EA) Remainder: width(mm)length(m)number(EA)	✓ Optimization Complete! Your steel cutting plan has been optimized for maximum efficiency. Optimized Output Width (mm) Standard Length (m) Bars Needed (EA) Total Cuts 2 12 11 11 Remainder Material Width (mm) Length (m) Pieces (EA) 2 100 11 Export Results
Impact - 'Why'	Header Title: Why Optimization Matters Subtitle: The business and environmental impact for smarter steel use Part A - What is Optimization? Steel optimization is the process of reducing debris, from human calculation, by finding the most efficient cutting pattern using complex mathematical algorithms. Part B - Benefits • Cost Savings: less material used means lower costs for raw materials • Environmental Impact: reducing steel waste leads to lowering the usage of energy and CO2 emissions which supports the sustainability goals • Structural Efficiency: Optimized designs are much more lighter, easier to build, and often last longer Section C - Real-World Target The optimization program aims to reach ≥97% accuracy, aiming for a waste of 1-2%.	SteelOptimize HomeCalculatorHow to UseImpact Why Optimization Matters The business and environmental impact for smarter steel use What is Optimization? Steel optimization is the process of reducing debris, from human calculation, by finding the most efficient cutting pattern using complex mathematical algorithms. Benefits Cost Savings Less material used means lower costs for raw materials Expected Impact 15-25% reduction in material costs Environmental Impact Reducing steel waste leads to lowering the usage of energy and CO₂ emissions which supports the sustainability goals Expected Impact 30% reduction in waste Structural Efficiency Optimized designs are much more lighter, easier to build, and often last longer Expected Impact 20% improved efficiency

Real-World Target

The optimization program aims to reach ≥97% accuracy, aiming for a waste of 1-2%.

Current Accuracy

97.2%

≥97%

Target Accuracy

1-2%

Target Waste

98.5%

Material Usage

24/7

Availability

Ready to Optimize Your Steel Usage?

Start reducing waste and saving costs with our advanced optimization algorithms

Start Optimizing Now